

MAXVQA: A Large-Scale Multi-Source Dataset for Explanatory Visual Question Answering in Mammography

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Abstract—Visual Question Answering (VQA) for medical imaging has focused predominantly on closed-ended classification, limiting its utility for the clinical explanations essential to trustworthy AI-assisted diagnosis. We present MAXVQA, a dataset of 125,591 question-answer pairs derived from 26,900 annotations across four mammography databases: CBIS-DDSM, VinDr-Mammo, INbreast and TOMPEI-CMMD. Unlike mammography VQA datasets that elicit single-word responses, MAXVQA emphasises *explanatory* answers across nine question types, including assessment reasoning, feature significance, counterfactual analysis, differential diagnosis and management. We introduce a unified annotation schema grounded in ACR BI-RADS and aligned with FDA MQSA requirements, an adaptive generation pipeline calibrated to each source's feature granularity, a ROUGE-L similarity filter, and an optional question-paraphrasing module. An information-content audit shows that, because answers are generated deterministically from templates, the 125,591 pairs contain 5,163 distinct answers; we therefore characterise our fine-tuning results as showing that a model learns to *reproduce* a schema-grounded explanation style, not as evidence of transferable clinical reasoning. To test visual grounding rather than assume it, we fine-tune a multimodal model under a controlled ablation ladder with a mismatched-image control, scoring recovery of *ground-truth clinical attributes* rather than template wording, so that memorising answer strings confers no advantage. A model trained on MAXVQA recovers ACR breast density from the mammogram in 66.1% of cases, against 25.9% with no image and a 41.1% majority-class baseline; it does not learn to determine malignancy, and we make no such claim. Evaluation against radiologist-authored free-text answers remains future work.

Index Terms—Visual Question Answering, Mammography, Explainable AI, Medical Imaging, BI-RADS, Breast Cancer Screening

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I. INTRODUCTION

Breast cancer is the most commonly diagnosed malignancy among women worldwide, and mammography the primary imaging tool for early detection and risk stratification [1]. Interpretation is challenging, reliant on radiologist expertise and susceptible to inter-reader variability [2]. As screening volumes grow, artificial intelligence (AI) has emerged as a promising adjunct, with prospective evaluations showing AI-assisted reading can match the diagnostic performance of expert radiologists [3], [4].

Yet current AI systems remain misaligned with how radiologists reason. Radiologists assess breast tissue composition, characterize abnormalities using standardized BI-RADS descriptors, correlate findings across ipsilateral and contralateral views, integrate temporal context from priors, and synthesize the evidence into a risk-stratified management recommendation. Most AI models bypass these steps, mapping raw pixels directly to diagnostic labels with no explicit intermediate reasoning, opening a gap between experimental performance and clinical utility, where transparent and verifiable reasoning is essential for trust and safe integration.

Several directions address this gap [5]: multi-view architectures encoding bilateral and ipsilateral view correspondences [6], [7]; vision-language models (VLMs) aligning mammographic images with clinical report text to enrich visual representations with radiological semantics [8], [9]; and Visual Question Answering (VQA) frameworks decomposing diagnosis into discrete natural language questions that encourage structured, interpretable intermediate outputs [10].

Each remains limited. Multi-view models are primarily discriminative, classifying without morphological explanation. VLMs treat clinical text as weak supervision for representation learning rather than a vehicle for structured reasoning, and cover only a narrow subset of finding types. Mammography VQA methods are predominantly closed-ended, restricted to questions such as “Is there a mass present?” or “What is the BI-RADS category?”, eliciting single-word responses that offer no insight into clinical reasoning; they omit abnormality characterization, feature-level descriptions, differential diagnoses, and the integrative reasoning underpinning trustworthy clinical decision support. When a radiologist meets a suspicious mammographic finding, the relevant questions extend far beyond binary classification:

- Why is this classified as BI-RADS 4C rather than BI-

RADS 3? What specific features drive this assessment?

- *What* is the clinical significance of the spiculated margins observed?
- *Why* can this finding *not* be classified as a simple cyst, despite some overlapping features?
- *What* differential diagnoses should be considered given the morphological characteristics?
- *What* is the recommended management pathway based on the current assessment?

Underlying these limitations is a dataset problem. Public mammography datasets are too small, inconsistently annotated, and largely devoid of the structured reasoning artifacts, feature-level descriptions, differential diagnoses, management rationales, and temporal comparison statements, needed to train and evaluate reasoning-aware models [11]. Without them, supervising, evaluating or validating intermediate reasoning steps is infeasible, so reasoning-grounded datasets are a prerequisite for advancing clinical AI in mammography. We address that gap through the following contributions:

- 1) **A cross-source harmonisation schema.** A unified annotation representation reconciling four mammography databases whose protocols, field names, value encodings and feature granularity differ substantially, plus the normalisation logic each requires. This, rather than scale, is our principal technical contribution.
- 2) **Adaptive, per-source question-type calibration.** A generator that determines which question types an annotation can legitimately support and *declines to generate the rest* rather than emit under-specified answers. VinDr-Mammo, lacking morphological descriptors, is restricted to four of the nine types.
- 3) **A structured BI-RADS knowledge base** encoding the ACR Atlas lexicon, positive predictive values, differential mappings and counterfactual reference entities, from which answers are generated *deterministically*. No language model produces answer content, which bounds the hallucination surface though, as we discuss, not systematic error.
- 4) **A quality-control pipeline** combining a ROUGE-L near-duplicate filter with automated checks including FDA MQSA management-protocol consistency, plus an optional question-side paraphrasing module. Its limits: deduplication is scoped within an annotation, so cross-annotation redundancy survives by design, and paraphrasing expands questions only.
- 5) **An evaluation protocol that tests, rather than assumes, visual grounding**, comprising an information-content audit of the corpus, a controlled image-ablation ladder with a mismatched-image control, leave-one-dataset-out transfer, and a non-circular utility evaluation that scores recovered clinical attributes rather than template wording.

II. RELATED WORK

A. Multi-View & Multi-Task Models

Beyond multi-view architectures exploiting complementary imaging projections for anatomical reasoning [12], [13], multi-

task models bring AI closer to clinical decision-making: joint abnormality localization and cancer classification [14], cancer classification with radiologist recall prediction [15], and cancer classification with BI-RADS assessment [6]. These improve efficiency and reinforce learning across related objectives but remain confined to a narrow, predetermined set of output categories. Defined at the label level, they never require natural language characterization of findings, explanation of an assessment, or engagement with the morphological attributes driving clinical decisions.

Vision–language models align mammographic images with clinical report text. CLIP-based and contrastive frameworks learn multimodal embeddings capturing visual tissue characteristics alongside radiological semantics [8], [16], using synthetic reports from metadata when paired text is unavailable [9]. Despite this richer signal, mammography VLMs target final diagnostic accuracy rather than structured reasoning: reports are weak supervision for representation learning, and the model need not generate or explain intermediate observations at inference. They also cover a narrow subset of finding types, predominantly masses and calcifications, where detection takes precedence over characterization: a model may identify a suspicious mass yet fail to describe its shape, margin, or density, precisely the attributes that anchor BI-RADS reasoning in practice.

B. Explainable AI

Post-hoc explainability methods include attention visualization [17], gradient-based saliency [18], and SHAP values [19]; in mammography they highlight the image regions that most influence a prediction [20]. Such explanations describe where a model attended rather than how it reasoned, and can appear clinically plausible while failing to faithfully reflect its true computational process [21], [22]. A heatmap may highlight a suspicious mass while the prediction was in fact driven by background texture or dataset-specific artifacts, an illusion of interpretability without clinical validity [23], [24]. Critically, such methods cannot reliably distinguish models that have learned genuine causal relationships from those exploiting spurious correlations.

An alternative paradigm builds *intrinsically interpretable* systems whose explanations are a byproduct of reasoning: prototype-based explanations [25], concept bottleneck models [26], and retrieval-augmented approaches [27]. Natural language generation is particularly promising, enabling explanations in the radiologist's professional vocabulary [28], [29].

C. Medical Visual Question Answering

Medical VQA is a promising path toward interpretable clinical AI. VQA-RAD [30] pioneered it with 3,515 question-answer pairs across radiology images, benchmarking modality, plane, organ system, and abnormality. VQA-Med [31] extended this through the ImageCLEF challenge series, and SLAKE [32] added 14,028 bilingual (English-Chinese) QA pairs with semantic labels enabling knowledge-enhanced reasoning. Later efforts scaled up: PathVQA [33] provides 32,799 pathology questions and PMC-VQA [34] 227,000 pairs

from PubMed Central figures. These general medical datasets have limited mammography coverage and are predominantly closed-ended, eliciting single-word or categorical responses that encode a final decision without exposing its justification, a structure that does not support the interrogation characterizing radiological practice.

MammoVQA [10] targets mammography directly, with 4,412 images and approximately 15,000 closed-ended QA pairs (Table I) spanning lesion identification, characterization, and BI-RADS assessment. It is an important advance, but its authors explicitly acknowledge that “*the classification-based design limits answers to closed categories, restricting LVLs’ open-ended reasoning potential.*” Beyond that constraint, mammography VQA models cover a restricted range of finding types and question dimensions, with limited attention to abnormality subtype characterization and virtually no integrative reasoning. We address this acknowledged gap with explanatory answers that articulate clinical reasoning across nine question types, spanning abnormality characterization, differential diagnosis, management recommendations, and evidence integration, grounded in ACR BI-RADS Atlas standards.

A useful contrast is ViX-Ray [35], a Vietnamese chest radiograph resource making almost the opposite design choices. Where MAXVQA optimises for scale, multi-source harmonisation and regulatory grounding by templating structured annotations, ViX-Ray is built on radiologist-written findings and evaluated against those human references. The trade-off is exact: templates buy consistency, volume and a bounded hallucination surface, at the cost of answer diversity and of references a model can reproduce without reading the image; human authorship buys references no template can predict, at the cost of scale. Sec. IV-F.3 adopts the latter’s guarantee by scoring against *ground-truth clinical attributes* rather than our template wording; a clinician-authored free-text test set remains future work (Sec. IV-H). Table I compares the MAXVQA with existing medical datasets utilized in the previous research.

III. METHODOLOGY

We describe the public mammography datasets, the MAXVQA generation pipeline, and the multi-agent LLM and radiologist validation methods.

A. Available Mammography Datasets

Several mammography datasets are publicly available [11]. CBIS-DDSM [36] provides 2,620 scanned film cases annotated with mass shape, margins, calcification morphology and distribution, BI-RADS assessment, and biopsy-proven pathology. VinDr-Mammo [37] provides 5,000 full-field digital exams (20,000 images) with breast-level BI-RADS assessments, density classifications, and finding-level bounding boxes. INbreast [38] provides 410 full-field digital images with precise lesion segmentation masks and BI-RADS assessments.

TOMPEI-CMMD [39], [40] provides 1,775 patients with breast-level BI-RADS assessments, density classifications, and abnormality type annotations (mass, calcification, asymmetry),

covering an East Asian screening population that complements the Western and Southeast Asian demographics of the other three sources. Each dataset follows its own protocol, and this heterogeneity is a significant barrier to direct cross-dataset adaptation. The scope of clinical information also varies considerably: CBIS-DDSM characterizes masses and calcifications in detail, including mass margin, mass shape, calcification distribution and morphology, whereas VinDr lacks such granular descriptors. This disparity in annotation richness makes the reasoning-based question-answer generation process differ between datasets.

B. Dataset Generation Pipeline

The MAXVQA pipeline takes heterogeneous annotations from four publicly available mammography databases, a structured BI-RADS knowledge base, and dataset-specific adaptive generation guidance, and produces clinically grounded explanatory question-answer pairs through normalization, generation, filtering, and validation. Fig. 1 shows the MAXVQA generation pipeline; Fig. 3 traces a source mammographic annotation end-to-end to a final QA pair.

1) *Unified Annotation Schema*: The four source databases underlying MAXVQA, CBIS-DDSM, VinDr-Mammo, INbreast, and TOMPEI-CMMD, differ substantially in annotation format, field naming, value encodings, and clinical feature granularity. To harmonize them we define a *UnifiedAnnotation* schema (Table II) giving a common representation regardless of origin. Fields needed for identification and provenance are required of all sources; clinical feature fields are optional, since availability varies by dataset. Every annotation is therefore processable by the adaptive QA generator regardless of feature completeness, the generator deriving applicable question types from which optional fields are populated.

Each loader maps source field names and value encodings to canonical schema forms. Normalization covers standardization to canonical BI-RADS lexicon terms, compound-value handling selecting the most clinically significant descriptor, BI-RADS category parsing from varied source formats, and conversion of ACR density letter grades to numeric categories. The vocabulary follows the ACR BI-RADS Atlas [41], the lexicon mandated by the FDA’s Mammography Quality Standards Act (MQSA) [42]. Alignment ensures that generated explanations use professionally recognised terminology familiar to practising radiologists. We are careful about what this does *not* establish. MQSA governs reporting terminology and management-recommendation protocols; it is not a validation regime for clinical decision support. That our answers use MQSA-mandated vocabulary and follow its management logic says nothing about their clinical safety, regulatory validity, or deployment readiness. MAXVQA is a research resource for developing and evaluating knowledge-grounded explanatory systems; any model trained on it would require independent clinical and regulatory validation before use.

2) *BI-RADS Knowledge Base*: Clinically accurate explanations rest on our structured encoding of the BI-RADS Atlas [41], which has six components: definitions for BI-RADS categories 0–6 with positive predictive values (PPV),

TABLE I

COMPARISON WITH EXISTING MEDICAL VQA DATASETS. MAXVQA UNIQUELY COMBINES LARGE SCALE, EXPLANATORY ANSWERS, MULTI-SOURCE INTEGRATION, AND MAMMOGRAPHY-SPECIFIC DEPTH.

Dataset	QA Pairs	Sources	Modality	Q Types	Answer	Explan.	Paraph.	Primary Focus
VQA-RAD [30]	3,515	1	Radiology	11	Closed	No	No	Modality, organ
VQA-Med [31]	15,292	1	Radiology	4	Closed	No	No	Abnormality
SLAKE [32]	14,028	1	Radiology	3	Mixed	Partial	No	Knowledge
PathVQA [33]	32,799	1	Pathology	7	Mixed	Partial	No	Location
PMC-VQA [34]	227,000	1	Mixed	Multiple	Mixed	No	No	Figures
MammoVQA [10]	~15,000	1	Mammo	4	Closed	No	No	Classification
MAXVQA	125,591	4	Mammo	9	Open	Yes	Opt.	Clinical reasoning

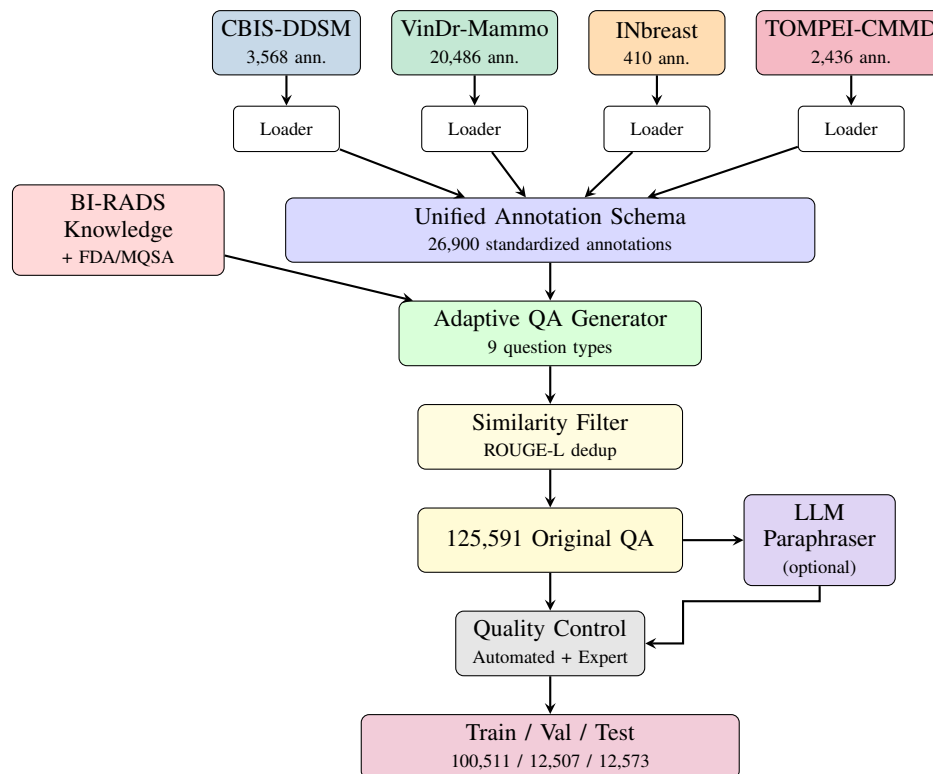


Fig. 1. MAXVQA generation pipeline. Four source datasets are normalized into a unified annotation schema; the adaptive QA generator, informed by a BI-RADS/MQSA knowledge base, produces QA pairs that are deduplicated by a ROUGE-L filter, optionally paraphrased, and validated before patient-level stratified splitting.

management recommendations and explanatory text; a mass shape lexicon over the four shape descriptors (oval, round, lobular, irregular) with clinical descriptions, suspicion levels and typical BI-RADS associations; a mass margin lexicon over the five margin descriptors (circumscribed, obscured, microlobulated, indistinct, spiculated) with malignancy PPV and suspicion-level reasoning; calcification descriptors spanning seven morphology types and five distribution patterns with suspicion levels and reasoning; differential diagnosis mappings linking feature combinations to likely diagnoses with PPV estimates; and counterfactual entries characterizing five reference entities, cyst, fibroadenoma, fat necrosis, benign lesion and invasive carcinoma, supporting contrastive reasoning about why a finding can or cannot be a given entity.

3) Adaptive Question Generation: Generation is *adaptive*: question types are calibrated to the feature granularity of each annotation. Each type carries specific feature requirements: descriptive needs an abnormality type or BI-RADS category; assessment reasoning and management need BI-RADS; density explanation needs breast density; differential diagnosis needs an abnormality type; the three feature-significance types need the corresponding descriptor (mass shape, mass margins, or calcification morphology and distribution) present in the knowledge base; and counterfactual questions need detailed features with BI-RADS ≥ 4 . The generator checks these per annotation and produces only the types whose required features are present. We additionally restrict types where limited annotations would yield generic, low-value answers. VinDr-Mammo lacks detailed abnormality characterization, so

TABLE II

UNIFIED ANNOTATION SCHEMA. FIELDS MARKED † ARE REQUIRED FOR ALL SOURCE DATASETS (IDENTIFICATION AND PROVENANCE); OPTIONAL FIELDS DEPEND ON SOURCE ANNOTATION GRANULARITY, AND THE ADAPTIVE GENERATOR PRODUCES QUESTION TYPES ONLY FOR FEATURES THAT ARE PRESENT.

Group	Field	Type / values
Identifiers	annotation_id†	string, unique identifier
	dataset_source†	string, source dataset
	patient_id†	string, patient/study
Location	laterality	LEFT, RIGHT
	view	CC, MLO
Finding	abnormality_type	mass, calcification, asymmetry, architectural_distortion
Mass	mass_shape	OVAL, ROUND, LOBULAR, IRREGULAR
	mass_margins	CIRCUMSCRIBED, OBSCURED, MICROLOBULATED, INDISTINCT, SPICULATED
Calcif.	calc_morphology	AMORPHOUS, FINE_PLEOMORPHIC, FINE_LINEAR_BRANCHING, etc.
	calc_distribution	DIFFUSE, REGIONAL, GROUPED, LINEAR, SEGMENTAL
Assessment	birads	int, 0–6
	breast_density	int, 1–4 (ACR)
	pathology	MALIGNANT, BENIGN

feature significance, counterfactual or differential diagnosis questions would be repetitive and under-specified; a differential diagnosis, for example, could only list generic possibilities without reference to specific morphological features. VinDr-Mammo is accordingly restricted to four types: descriptive, assessment reasoning, density explanation, and management.

4) *Similarity Filtering*: Template-based generation can produce identical or near-identical answers when annotations share the same clinical features, for example all BI-RADS 4 management answers recommending tissue sampling. To control this redundancy without discarding answers legitimately identical across different images, we apply a two-stage similarity filter scoped at the annotation level, comparing only pairs that share an annotation ID and question type. Exact deduplication first collapses content-identical answers within each group to the most metadata-rich instance. Near-duplicate detection then flags answers similar enough to be redundant: a trigram Jaccard pre-filter (threshold ≥ 0.40) identifies candidate pairs, scored with ROUGE-L F1; pairs exceeding a similarity threshold of 0.70 are deduplicated, again retaining the richer instance. This scoping is intentional: two different mammographic images with the same BI-RADS category may correctly share identical answer text, and comparing across images would incorrectly remove these valid pairs.

5) *Where Language Models Are, and Are Not, Used*: Reviewer 3 asks how hallucinations and clinically incorrect outputs were detected and corrected during generation. Answering requires a clarification the original manuscript left implicit, which materially changes the pipeline's risk profile: **no language model generates answer content in MAXVQA**. Every answer is assembled deterministically from the BI-RADS knowledge base by templates whose slots are filled from normalised annotation fields. A language model is used at exactly one point, paraphrasing *questions* (Sec. III-B.6), with outputs filtered before acceptance; answers are never

rewritten by a model. The free-form fabrication characteristic of LLM-generated corpora therefore has essentially no surface on which to occur: a template cannot invent a finding that is not in the annotation.

We resist a stronger claim. Determinism protects against fabrication but not against *systematic* error. If a knowledge-base entry encodes an incorrect positive predictive value, or a template licenses an inference a radiologist would not make, that error is reproduced identically across every pair it touches, and, given the redundancy documented in Sec. IV-B, potentially across tens of thousands of them. Our exposure is not hallucination but correlated, high-multiplicity error: precisely the failure mode automated checks are least able to detect and expert human review most needed to catch. This is the principal argument for expanding the radiologist panel rather than scaling the LLM judge.

6) *LLM-Based Paraphrasing & Quality Control*: To increase linguistic diversity, the pipeline optionally paraphrases questions with a language model, supporting Ollama (LLaMA 3.1 8B or Mistral 7B), HuggingFace 4-bit inference, and a rule-based fallback, generating two variations per question. Paraphrases are filtered: they must end in “?”, preamble is stripped, a 15-character minimum is enforced, and duplicates are discarded. Only questions are paraphrased; answers are left unchanged, preventing factual drift. Applied in full, paraphrasing expands the dataset by approximately $2.98\times$.

All QA pairs, original or paraphrased, then undergo automated quality control of six validation checks: completeness, that required fields are present and non-empty; length, that answers fall between 50 and 2,000 characters, ensuring substantive responses; placeholder detection, that no unfilled template tokens remain; and terminology validation, that question-type-specific clinical terms are present (e.g. “margin” or “shape” in feature-specific answering) with no preamble. Quality control further checks alignment with the FDA MQSA framework [42], verifying that management recommendations follow mandated reporting requirements: BI-RADS 0 recommends additional imaging, BI-RADS 3 short-interval follow-up, and BI-RADS 4–5 tissue diagnosis. No generated explanation may deviate from these established clinical protocols.

C. Dataset Validation

We additionally validate MAXVQA by multi-agent LLM evaluation with specialized radiologist personas, independent expert evaluation by two senior consultant radiologists, and baseline text-generation experiments assessing its utility for improving clinical reasoning in language models.

1) *Multi-Agent LLM*: To assess clinical quality at scale, three LLM-based evaluator agents, each with a distinct radiologist persona, independently rate a stratified sample of generated QA pairs. This follows recent work on LLM-as-judge evaluation [43]. We originally stated that persona diversification mitigates single-evaluator bias. It does not: three personas sharing one backbone are not independent evaluators, and prompting a model to adopt a role does not decorrelate its errors. Because the same LLaMA 3.1 8B model is also the generator and the fine-tuning target in Sec. IV-E, the arrangement further invites

model-family bias, whereby text in a style native to LLaMA is both favourably judged by LLaMA and easily imitated by it. The revised protocol therefore draws judges from different model families and treats the human evaluation as primary. We retain the description below because it documents what was actually run. We sample 200 QA pairs from the test set (approximately 22 per question type), balanced across BI-RADS categories and source datasets. The three agents use LLaMA 3.1 8B [44] via Ollama with distinct system prompts:

- *Senior Radiologist*: 15+ years of mammography experience; prioritizes clinical accuracy, BI-RADS compliance, and FDA MQSA alignment. Flags any claims that could affect patient safety.
- *Academic Radiologist*: Research and education focus; evaluates reasoning depth, educational value, and evidence-based content. Assesses whether a trainee would learn correct reasoning from the answer.
- *Clinical Radiologist*: Daily breast imaging practice; evaluates practical utility, actionability of recommendations, and communication efficiency. Focuses on real-world workflow relevance.

Each agent independently rates every QA pair on three dimensions using a 5-point Likert scale:

- *Clinical Accuracy* (1–5): Are the factual claims correct per ACR BI-RADS standards?
- *Completeness* (1–5): Does the answer adequately address the question?
- *Clinical Relevance* (1–5): Would this explanation be useful in clinical practice or education?

Each agent receives the full clinical context (dataset source, abnormality type, BI-RADS category, pathology, morphological features, and breast density) with the question and answer. This has a consequence we did not previously acknowledge: the judge sees precisely the fields from which the answer was generated. It can verify only that a templated answer is *compatible with its own generating metadata*, a form of internal consistency, and cannot assess whether the answer is faithful to the mammogram. The LLM evaluation provides no evidence of image-faithfulness and we no longer imply that it does; image-faithfulness is addressed in Sec. IV-F. Low temperature ($T = 0.1$) ensures consistent scoring. Agents return structured JSON ratings with regex-based fallback parsing for robustness. All 600 evaluations (200 QA pairs $\times$ 3 agents) completed with zero parsing failures.

2) Human Radiologist: Validating the clinical content of MAXVQA ultimately requires expert human judgment. Two senior consultant radiologists, each with more than 20 years of clinical mammography experience, independently evaluated a stratified random sample of 100 QA pairs from the test set (94 completed by both raters; see Sec. IV-D), balanced across all nine question types and three source datasets (CBIS-DDSM, TOMPEI-CMMD, VinDr-Mammo).

Each radiologist rated every QA pair on four dimensions using a 5-point Likert scale:

- *Clinical Accuracy* (1–5): Are the factual claims correct per ACR BI-RADS standards?

- *Completeness* (1–5): Does the answer adequately address the question?
- *Clinical Relevance* (1–5): Would this explanation be useful in clinical practice or radiology education?
- *Visual Consistency* (1–5): Does the textual description match the mammographic image?

A custom web interface (Streamlit) presented the clinical context, question, generated answer, and mammographic images where available. Evaluators were blinded to each other's ratings and worked independently. Inter-rater reliability used Cohen's quadratic-weighted kappa (κ_w) and ICC(2,1) for absolute agreement.

3) Clinical Quality Benchmarking: We evaluate LLaMA 3.1 8B [44] in three text-only settings on stratified test samples, supplying clinical context as structured metadata throughout to isolate reasoning from visual perception. In the **zero-shot** setting the model receives only the clinical context, abnormality type, BI-RADS category, mass features, breast density, and pathology, plus the question and a system prompt instructing a detailed clinical explanation in standard ACR BI-RADS terminology.

The **3-shot** setting adds three question-type-matched training examples, demonstrating the expected answer structure and clinical depth without parameter updates. Both settings use 500 stratified test samples (approximately 55 per question type) at temperature $T = 0.3$. The **QLoRA fine-tuned** setting tunes LLaMA 3.1 8B Instruct with 4-bit QLoRA [45], rank 32, alpha 64, and LoRA adapters on all attention and MLP projection layers, training on 1,000 stratified samples for 3 epochs with a cosine schedule (peak 2×10^{-4} , effective batch size 16). This deliberately constrained scenario, $<1\%$ of the training set, establishes a lower bound on fine-tuned performance rather than an optimized ceiling. Evaluation uses greedy decoding on the same 500 stratified test samples.

We report four metrics: BLEU-4 [46], n-gram precision against reference answers; ROUGE-L, longest-common-subsequence overlap; BERTScore F1 [47], semantic similarity; and Clinical Keyword Recall (CKR), the fraction of question-type-specific clinical terms in the reference answer preserved in the generated output. Following Reviewer 1, BERTScore uses RoBERTa-large with baseline rescaling for all vision-grounded results (Sec. IV-F), while the text-only baselines retained for continuity use the original configuration. We also corrected an earlier CKR that returned 1.0 when the reference contained no keyword, expanded its keyword lists, and use the corrected metric for all vision-grounded results. Together these capture surface fidelity and clinical relevance.

D. Pipeline Faithfulness

Because every answer is generated deterministically from the BI-RADS knowledge base, the pipeline can be validated against that source. We checked three invariants across all 125,591 QA pairs: the BI-RADS category stated in an answer matches the annotation in 100.0% of the 102,849 answers stating one; a management answer's recommendation matches the knowledge-base management for the assigned category in 100.0% of 26,893 cases; and every one of the 4,560

counterfactual answers stating a category invokes one other than the annotation's, confirming these items reason counterfactually rather than restate the assessment. Complete concordance is the intended behaviour of deterministic generation: the pipeline introduces no additional error. It bounds the failure mode accordingly: the exposure is not hallucination but correlated systematic error, in which one incorrect knowledge-base entry propagates identically across every pair it touches. This validation concerns faithfulness to the knowledge base and does not certify the base's own clinical accuracy.

IV. RESULTS

A. Dataset Statistics

MAXVQA comprises 125,591 question-answer pairs from 26,900 annotations across four publicly available mammography databases, spanning nine clinically grounded question types and 8,673 unique patients. Table III reports the per-source breakdown before and after paraphrasing with patient-level stratified splits. The source distribution reflects the relative scale and annotation depth of the original collections: CBIS-DDSM and TOMPEI-CMMD support all nine question types through their full morphological and pathological annotations, while VinDr-Mammo and INbreast are restricted to four and five. Despite that richer depth, VinDr-Mammo dominates by volume (65.2%) through its far larger annotation count, mirroring the availability landscape of public mammography data. With the optional paraphrasing module the dataset expands to 374,261 pairs, splits scaling proportionally.

Table III reports the nine question types, which divide into universal and feature-dependent categories. The universal types, descriptive, assessment reasoning, management, and density explanation, are generated for all annotations and account for 85.6% of the dataset, roughly 21.4% each. The feature-dependent types, differential diagnosis, counterfactual, margin, shape and calcification significance, require specific morphological descriptors and are less frequent, from 4.4% for counterfactual pairs down to 1.5% for calcification significance. This imbalance is not an artifact of pipeline design but reflects the annotation landscape: detailed morphological descriptors appear in only a subset of the source collections, and their scarcity here mirrors their uneven availability across clinical databases. It highlights a broader limitation of existing public mammography resources and reinforces the need for richer, reasoning-oriented annotation in future data collection.

We foreground this imbalance because it sits in tension with how we motivate the dataset. MAXVQA is presented as a resource for *explanatory reasoning*, yet 85.6% of its mass is the four universal types, descriptive, assessment, management and density, its most formulaic content. The reasoning-heavy types that carry the argument of this paper are a thin tail: counterfactual 4.4%, differential 4.3%, mass margin 2.1%, mass shape 2.1%, calcification 1.5%. That tail is also the most redundant part of the corpus (Sec. IV-B). Any figure averaged over the nine types is therefore dominated by the formulaic majority and tells the reader little; we report per-type results throughout for that reason.

B. Nominal Versus Effective Dataset Size

Because answers are generated deterministically from a finite template set, the number of QA pairs overstates the number of distinct explanations. Reviewer 1 asked us to quantify this, and we do so without minimisation, since it is also the mechanism behind the fine-tuning behaviour reported in Sec. IV-E.

Table IV reports, per question type, the number of QA pairs, distinct answer strings, entropy H of the empirical answer distribution, and effective number of distinct answers 2^H . Across the corpus, 125,591 QA pairs contain **5,163 distinct answers** (4.1%), with an answer entropy of 8.49 bits, an effective vocabulary of roughly 360 equiprobable answers. The three feature-significance types are near-degenerate: 2,644 mass-margin pairs are drawn from *five* distinct answers, 2,617 mass-shape pairs from five, and 1,871 calcification pairs from six. Per source, VinDr-Mammo, 65.2% of the corpus, contributes 81,944 pairs drawn from 636 distinct answers.

Two consequences follow.

First, the fine-tuning result in Sec. IV-E is explained by this table. A BLEU-4 of 1.0000 on the feature-significance types is a model reproducing five strings. It is not evidence of clinical reasoning and we do not present it as such.

Second, structured metadata renders the mammogram informationally redundant. For four of the nine question types, differential diagnosis and all three feature-significance types, the answer is an *exact* deterministic function of the annotation's metadata tuple: across all 125,591 pairs, no metadata tuple maps to more than one answer. For the remaining five types the answer is determined by metadata up to at most eight stylistic variants (sentence-opener selection, keyed to a hash of the annotation identifier, and optional laterality mentions), none of which carry clinical information. It follows that a model given the structured findings can score highly *without ever consulting the image*, and that conditions (b) and (d) of the ablation ladder in Sec. IV-F cannot be separated by visual evidence. This is a property of how the corpus was built; it was true of the experiments in our original submission; and it is precisely why those experiments could not have tested visual grounding.

The redundancy is partly by design: two mammograms exhibiting spiculated margins *should* attract the same explanation of what spiculated margins signify, and the within-annotation scoping of our similarity filter (Sec. III-B.4) deliberately preserves that. The distinction between a corpus of 125,591 instances and one of 5,163 explanations is nonetheless material to anyone judging what the resource is worth, and we report it prominently.

C. Multi-Agent LLM Evaluation

Table VI presents the multi-agent evaluation across 600 assessments. A re-analysis of the raw judgements forces a caveat, which we state before the scores themselves.

The judge does not use the rating scale. Across all 600 judgements the LLaMA 3.1 8B evaluator assigned clinical-accuracy scores of only 4 or 5, 545 fours and 55 fives; no persona ever assigned a 1, 2 or 3 to any item. Clinical

TABLE III

DATASET COMPOSITION. **TOP:** STATISTICS BEFORE AND AFTER PARAPHRASING; Q TYPES IS THE NUMBER OF APPLICABLE QUESTION TYPES PER SOURCE AND SPLIT RATIOS (80/10/10) ARE APPLIED AT THE PATIENT LEVEL TO PREVENT LEAKAGE. SPLIT COUNTS CORRECTED AGAINST THE RELEASED SPLIT FILES. **BOTTOM:** QUESTION TYPE DISTRIBUTION; UNIVERSAL TYPES ARE GENERATED FOR ALL ANNOTATIONS, WHILE FEATURE-DEPENDENT TYPES REQUIRE SPECIFIC MORPHOLOGICAL DESCRIPTORS.

Dataset	Annotations	%	Q Types	Annotation Depth	Original (post-filtering)			After Paraphrasing		
					QA Pairs	Train	Val/Test	QA Pairs	Train	Val/Test
VinDr-Mammo	20,486	65.2	4	BI-RADS, density	81,944	65,452	8,280/8,212	244,193	195,354	24,420/24,419
CBIS-DDSM	3,568	20.8	9	Full morphology + pathology	26,139	21,037	2,497/2,605	77,894	62,315	7,790/7,789
TOMPEI-CMMD	2,436	12.3	9	Full morphology + pathology	15,459	12,383	1,525/1,551	46,068	36,854	4,607/4,607
INbreast	410	1.6	5	BI-RADS, abnormality type	2,049	1,639	205/205	6,106	4,885	611/610
Total	26,900	100			125,591	100,511	12,507/12,573	374,261	299,408	37,428/37,425

Universal Types					Feature-Dependent Types					Total
	Descriptive	Assessment	Management	Density	Differential	Counterfactual	Mass Margin	Mass Shape	Calcification	
Count	26,900	26,893	26,893	26,897	5,363	5,513	2,644	2,617	1,871	125,591
%	21.4	21.4	21.4	21.4	4.3	4.4	2.1	2.1	1.5	100

TABLE IV

NOMINAL VERSUS EFFECTIVE SIZE OF MAXVQA. H IS THE ENTROPY OF THE EMPIRICAL ANSWER DISTRIBUTION AND 2^H THE EFFECTIVE NUMBER OF DISTINCT ANSWERS.

Question type	Pairs	Unique	Ratio	H (bits)	2^H
Descriptive	26,900	3,885	0.144	9.03	522
Density explanation	26,897	82	0.003	5.34	41
Management	26,893	76	0.003	4.92	30
Assessment reasoning	26,893	1,008	0.037	5.91	60
Counterfactual	5,513	61	0.011	4.32	20
Differential diagnosis	5,363	35	0.007	3.94	15
Feat. sig. (margins)	2,644	5	0.002	2.17	4
Feat. sig. (shape)	2,617	5	0.002	1.94	4
Feat. sig. (calcification)	1,871	6	0.003	2.01	4
Total	125,591	5,163	0.041	8.49	360

relevance is likewise confined to $\{4, 5\}$ (241 fours, 359 fives) and completeness to $\{3, 4, 5\}$ (20, 47 and 533 respectively). The mean clinical accuracy of 4.09 therefore summarises not a distribution of quality judgements but a judge that answered “4” 91% of the time. We do not treat the LLM accuracy dimension as a measurement of clinical accuracy, and we no longer offer it as a headline validation of the dataset. An 8B model is, as Reviewer 1 observes, a weak instrument for adjudicating clinical accuracy, and persona diversification cannot correct this because all three personas share one backbone. The human evaluation (Sec. IV-D) is accordingly our primary evidence of clinical quality; the LLM evaluation is retained only as a scalable secondary screen.

We also correct an error of our own. The original submission reported a single inter-agent Fleiss’ κ of 0.440. No such statistic exists in our results: the analysis records three per-dimension values, accuracy 0.560, completeness 0.272, relevance 0.487, of which 0.440 is the undisclosed unweighted mean. Table VI now reports all three.

1) *A multi-family judge panel:* To test whether the scale collapse is a property of the task or of the judge, we reran the evaluation with a panel of four judges from distinct model families (Qwen2.5-14B, Llama-3.1-8B, Gemma-2-9B and Mistral-7B), each scoring the identical 198-item sample,

792 judgements with no parse failures. The panel settles the question in the reviewers’ favour and locates the cause in the original judge rather than the benchmark: three of four judges discriminate across the rating scale, and two (Qwen2.5-14B and Gemma-2-9B) use all five accuracy points, so the original judge’s confinement to $\{4, 5\}$ reflected that model, not an absence of gradation to detect. We report every judge’s full score distribution rather than a mean (Table V), and subject each judge to an automatic degeneracy check whose verdict we report in full: Mistral-7B collapses on completeness and relevance, assigning the maximum score to 98% and 97% of items, reproducing the exact failure of our original evaluator; we flag those dimensions as unusable and exclude them rather than average them into any summary. Among the usable judges, cross-family agreement is modest and positive (mean pairwise Spearman $\rho = 0.32$ for relevance, 0.21 for accuracy, 0.14 for completeness). We draw no stronger conclusion than the data support: several independent judges agree weakly on the quality of templated answers, consistent with the limited discriminative content those answers carry (Sec. IV-B), and the reason we treat expert human review, not LLM scoring, as the primary evidence.

The six question types reporting 4.00 ± 0.00 in Table VI are not evidence of uniform quality; they are the signature of a judge returning a constant. We previously read the per-type pattern as clinically meaningful, attributing Management’s 4.56 to the MQSA-aligned pipeline and Counterfactual’s low relevance to the specialisation of contrastive reasoning. Given the degeneracy documented above, those readings are unsupported and we withdraw them. The only interpretable signal in the accuracy column is that the judge departed from its default of 4 on two question types, far too thin a basis for a claim about the dataset.

Per-dimension inter-agent agreement is moderate for accuracy ($\kappa = 0.560$) and relevance ($\kappa = 0.487$), and fair for completeness ($\kappa = 0.272$) [48]. These figures should not be read as reassurance: three personas of the same 8B model agreeing that an answer is a “4” is agreement about a default, not concordant clinical judgement. Genuine evaluator independence requires different model families, which the

TABLE V

MULTI-FAMILY LLM JUDGE PANEL: SCORE DISTRIBUTIONS ON 198 QA PAIRS (792 JUDGEMENTS, NO PARSE FAILURES). EACH CELL GIVES THE COUNT OF 1/2/3/4/5 RATINGS. SOURCE:

E3_JUDGE_PANEL.JSON.

Judge (family)	Dim.	1	2	3	4	5	mean
Qwen2.5-14B (Alibaba)	Acc.	2	12	42	127	15	3.71
	Comp.	0	15	111	22	50	3.54
	Rel.	0	3	47	109	39	3.93
Llama-3.1-8B (Meta)	Acc.	0	2	65	72	59	3.95
	Comp.	0	8	27	48	115	4.36
	Rel.	0	0	21	111	66	4.23
Gemma-2-9B (Google)	Acc.	3	3	33	25	134	4.43
	Comp.	0	8	22	150	18	3.90
	Rel.	0	5	41	104	48	3.98
Mistral-7B (Mistral AI)	Acc.	0	0	8	131	59	4.26
	Comp. [†]	0	0	0	4	194	4.98
	Rel. [†]	0	0	0	5	193	4.97

[†] flagged DEGENERATE ($\geq 97\%$ modal, ≤ 2 scale points); excluded.

TABLE VI

MULTI-AGENT LLM EVALUATION RESULTS (MEAN $\pm$ SD ON 5-POINT LIKERT SCALE, $n = 600$ EVALUATIONS ACROSS 3 AGENTS, MODEL: LLAMA 3.1 8B).

Question Type	Accuracy	Complete.	Relevance
Assessment Reasoning	4.24 $\pm$ 0.42	4.78 $\pm$ 0.53	4.68 $\pm$ 0.47
Counterfactual	4.00 $\pm$ 0.00	4.58 $\pm$ 0.78	4.30 $\pm$ 0.46
Feat. Sig. (Mass Shape)	4.00 $\pm$ 0.00	4.97 $\pm$ 0.17	4.77 $\pm$ 0.42
Feat. Sig. (Mass Margin)	4.00 $\pm$ 0.00	4.98 $\pm$ 0.12	4.74 $\pm$ 0.44
Feat. Sig. (Calcification)	4.00 $\pm$ 0.00	4.94 $\pm$ 0.24	4.65 $\pm$ 0.48
Differential Diagnosis	4.00 $\pm$ 0.00	4.89 $\pm$ 0.35	4.65 $\pm$ 0.48
Management	4.56 $\pm$ 0.50	4.71 $\pm$ 0.45	4.68 $\pm$ 0.47
Descriptive	4.00 $\pm$ 0.00	4.89 $\pm$ 0.43	4.14 $\pm$ 0.34
Density Explanation	4.02 $\pm$ 0.12	4.95 $\pm$ 0.21	4.76 $\pm$ 0.43
Overall	4.09 $\pm$ 0.29	4.86 $\pm$ 0.44	4.60 $\pm$ 0.49

Inter-agent Fleiss' κ (per dimension):
accuracy 0.560, completeness 0.272, relevance 0.487

revised protocol adopts.

D. Human Radiologist Evaluation

Two senior consultant radiologists, each with more than 20 years of mammography experience, independently assessed a stratified sample of QA pairs on the four dimensions defined above. Of the 100 items released for rating, both completed 94, and all statistics below use those 94 shared items. (The original submission reported $n = 90$, an interim snapshot taken before rating was complete; the final data are used here.)

Re-analysis undermines the way we previously summarised this evaluation. Per-rater means appear in Table VII; we no longer lead with a pooled score, for a reason the score distributions make plain. Rater R2 assigned the maximum score of 5 to 358 of 374 ratings (95%), scoring below 4 on only twelve occasions; rater R1 used the full scale (23 ones, 72 twos, 48 threes, 106 fours, 125 fives). A pooled mean across a discriminating rater and a near-constant one is not an informative summary of dataset quality, and the “overall

TABLE VII

HUMAN RADIOLOGIST EVALUATION ($n = 94$ QA PAIRS RATED BY BOTH RATERS). κ_w = QUADRATIC-WEIGHTED COHEN'S KAPPA; ICC = ICC(2,1); ρ = SPEARMAN RANK CORRELATION BETWEEN RATERS. NO POOLED MEAN IS REPORTED (SEE TEXT).

Dimension	R1	R2	κ_w	ICC(2,1)	ρ	p_ρ
Clin. Accuracy	3.29	4.97	0.013	0.014	0.087	0.406
Completeness	4.20	4.95	0.021	0.021	0.040	0.704
Clin. Relevance	3.73	4.95	0.028	0.029	0.143	0.168
Vis. Consistency	3.24	4.68	0.062	0.182	0.255	0.013

R1 scores {1:23, 2:72, 3:48, 4:106, 5:125}; R2 scores {1:0, 2:5, 3:7, 4:4, 5:358}

quality score of 4.26/5” previously reported as a headline is withdrawn.

We previously attributed the low κ_w and ICC values to a systematic difference in scoring stringency. That explanation does not survive scrutiny and we withdraw it. As Reviewer 1 observes, a pure additive offset preserves rank order and would not deflate a quadratic-weighted κ to 0.013; the hypothesis is testable, and we tested it. Spearman rank correlations between the two raters are $\rho = 0.087$ ($p = 0.41$) for clinical accuracy, $\rho = 0.040$ ($p = 0.70$) for completeness, and $\rho = 0.143$ ($p = 0.17$) for clinical relevance. The raters do not agree on item ordering either, and the offset account is refuted.

The cause is the rating behaviour above: a rater who assigns 5 to 96% of items conveys almost no information about which items are better, leaving little rank signal for the second rater to agree with. Bland–Altman analysis confirms a large positive bias (R2 minus R1: +1.68 on clinical accuracy) with wide limits of agreement ($[-0.84, +4.21]$): the disagreement is not a stable offset but a broad scatter.

One finding survives, and it is the most interesting result in this section. The only dimension on which the two raters share significant rank agreement is *visual consistency* ($\rho = 0.255$, $p = 0.013$; $\kappa_w = 0.062$, ICC = 0.182), the one dimension requiring them to look at the mammogram rather than assess the text. Where the raters had to consult evidence outside the generated answer, they began to agree; where they judged the prose itself, they did not. We take this as an early, if fragile, indication that image-grounded judgements are the ones worth collecting, and one reason the revised protocol foregrounds them.

A per-question-type breakdown rests on roughly ten items per type, is not statistically meaningful, and is therefore not reported.

Per-source means were closely comparable and are not broken out, since a few dozen items per source cannot support that comparison.

On the three shared dimensions the LLM and human means were clinical accuracy 4.09 versus 4.09, completeness 4.86 versus 4.54, and relevance 4.60 versus 4.34. We previously called the coincidence of the two clinical-accuracy means “exact agreement” and offered it as evidence that the multi-agent framework is a reliable proxy for expert assessment. That claim was wrong on two counts and is withdrawn.

First, as Reviewer 2 notes, equality of aggregate means is not agreement: two rating systems can share a mean and dis-

agree on every individual item. Second, and more seriously, the two evaluations were never run on the same items. The LLM evaluation sampled 200 QA pairs and the human evaluation 100, and the two samples intersect in *three* items. The means compared are computed over essentially disjoint sets, so their coincidence is arithmetic accident and carries no evidential weight whatever.

Item-level agreement between LLM and human judgements was therefore not merely unreported but *uncomputable* from the data we collected, since establishing it requires both evaluators to score an identical item set. We accordingly withdraw the “exact agreement” claim entirely, report the two sets of means above only as summaries of separately-drawn samples and not as paired evidence, and identify a shared-item agreement study as future work (Sec. IV-H).

E. Clinical Quality Benchmark Evaluation

We report these experiments under an explicit reinterpretation. They are *text-only*: the model receives structured clinical findings and never sees a mammogram, and as Sec. IV-B establishes, the answers are deterministic functions of exactly those findings. What follows therefore measures how readily a language model learns to reproduce a schema-grounded explanation style from metadata, a legitimate thing to measure, but not visual question answering and not transferable clinical reasoning. Every claim of “clinical reasoning” attached to these numbers in the original submission is withdrawn. The experiments that test visual grounding, evaluated against ground-truth labels rather than our templates, are reported in Sec. IV-F.

Table VIII presents results across the three settings and all question types, a clear progression: zero-shot LLaMA 3.1 8B as the unguided baseline, 3-shot in-context learning showing what the dataset teaches without parameter updates, and QLoRA fine-tuning on a constrained 1,000-sample subset as a lower bound on fine-tuned performance.

The zero-shot condition pairs high BERTScore (0.820) with very low BLEU-4 (0.036). A general-purpose LLM therefore holds substantial mammography domain knowledge, its semantic content broadly correct, but cannot produce the structured, BI-RADS-grounded explanations characterizing MAXVQA answers: it understands the concepts but lacks radiologists’ clinical communication patterns. Per-type CKR reveals important disparities. Assessment reasoning, calcification and margin feature significance achieve CKR of 1.0 without any examples, so the model produces these terms naturally when discussing mammography. Counterfactual reasoning reaches only 0.37 and management 0.50, showing that contrastive clinical reasoning and MQSA-specific management terminology are not part of its default generation. These are precisely the types where MAXVQA contributes most as a training resource.

Three examples per question type improve all metrics substantially: BLEU-4 nearly 10-fold ($0.036 \rightarrow 0.343$), ROUGE-L 2.3-fold ($0.204 \rightarrow 0.470$), and CKR from 0.749 to 0.935. Some types gain dramatically: counterfactual CKR rises from 0.37 to 0.96, a near-complete recovery of clinical terminology

through in-context learning alone. Descriptive and density explanation reach the highest absolute BLEU-4 here (0.528 and 0.584), reflecting a formulaic structure readily learned from few examples. Management CKR improves from 0.50 to 0.89, a substantial but incomplete gain, suggesting MQSA-aligned management protocols require more exposure than three examples provide. Differential diagnosis shows the weakest 3-shot response (BLEU-4 of 0.064), consistent with the variability of valid differentials that makes structural alignment from few examples difficult.

The QLoRA fine-tuned model, trained on 1,000 samples, reaches an overall BLEU-4 of 0.949 and attains *exactly* 1.0000 on BLEU-4, ROUGE-L and BERTScore for all three feature-significance types. We previously described this as the model having “completely internalized” those explanations. It is more accurate, and less flattering, to say it memorised them: those types contain five, five and six distinct answers respectively (Table IV). A score of exactly 1.0000 against a five-string reference set measures the reference set, not the model, and certainly not clinical reasoning.

Read correctly, the progression still tells us something, but something narrower than we claimed. Zero-shot shows a general-purpose model already possesses much of the relevant mammographic *semantics* (BERTScore 0.820) while producing almost none of the *form* (BLEU-4 0.036); three examples supply most of the form (BLEU-4 0.343), and a thousand training samples essentially all of it. The defensible conclusion is that MAXVQA teaches a schema-grounded explanation *style* very efficiently, and that the saturated rows mark where a templated reference set ceases to discriminate between models at all. That, rather than the headline BLEU-4, is why a non-circular evaluation is necessary; we provide one by scoring recovery of ground-truth clinical attributes from the image (Sec. IV-F.3), with a free-text human benchmark identified as future work (Sec. IV-H); and it is why we no longer describe the full-training-set result as “expected to further improve performance”. Against a reference set this degenerate, there is nothing left to improve.

Per-source results, requested by Reviewer 2 and recomputed from the released predictions without re-training, appear in Table IX and corroborate the concern about source-specific exploitation. In the 3-shot setting, BLEU-4 on VinDr-Mammo reaches 0.533, more than double CBIS-DDSM (0.234), unsurprising given Sec. IV-B: VinDr’s 81,944 pairs are drawn from just 636 distinct answers and are correspondingly easy to pattern-match from three examples. The source-balanced (macro) average is accordingly lower than the sample-weighted (micro) average reported originally (3-shot BLEU-4: 0.310 macro versus 0.343 micro). We report both, and treat the macro average as the more representative summary.

F. Vision-Grounded Baselines

No experiment in our original submission allowed the mammogram to participate; Reviewer 2 proposed the controlled comparison we adopt here: (a) question alone; (b) question with the structured findings, as in the original submission; (c) mammogram with question; (d) all three; and (e) a *mis-*

TABLE VIII

BASILINE VQA PERFORMANCE ON MAXVQA TEST SET ($n=500$ STRATIFIED SAMPLES, ≈ 55 PER QUESTION TYPE) ACROSS THREE EXPERIMENTAL SETTINGS. QLoRA FINE-TUNED USES 4-BIT LORA ADAPTERS TRAINED ON 1,000 SAMPLES ($< 1\%$ OF TRAINING SET). B-4 = BLEU-4, R-L = ROUGE-L, BS = BERTSCORE. BEST OVERALL SCORES IN **BOLD**.

Question Type	Zero-shot				3-shot				QLoRA fine-tuned			
	B-4	R-L	BS	CKR	B-4	R-L	BS	CKR	B-4	R-L	BS	CKR
Assessment Reasoning	0.0240	0.1920	0.8224	1.0000	0.2896	0.4104	0.8770	1.0000	0.8877	0.9248	0.9837	1.0000
Counterfactual	0.0248	0.1660	0.8149	0.3661	0.3639	0.5075	0.8958	0.9643	0.9730	0.9903	0.9957	1.0000
Density Explanation	0.0399	0.2181	0.8134	0.6250	0.5837	0.6669	0.9333	0.9345	0.9494	0.9661	0.9909	1.0000
Descriptive	0.0251	0.1988	0.8281	0.8750	0.5279	0.6531	0.9342	0.9167	0.8213	0.8723	0.9766	0.9940
Differential Diagnosis	0.0323	0.1897	0.8144	0.8000	0.0636	0.2168	0.8208	0.9333	0.9776	0.9834	0.9963	1.0000
Feat. Sig. (Calcif.)	0.0644	0.2108	0.8218	1.0000	0.2221	0.3550	0.8410	0.8929	1.0000	1.0000	1.0000	1.0000
Feat. Sig. (Mass Margin)	0.0482	0.2312	0.8331	1.0000	0.2862	0.4310	0.8789	1.0000	1.0000	1.0000	1.0000	1.0000
Feat. Sig. (Mass Shape)	0.0417	0.2347	0.8177	0.5727	0.2264	0.3683	0.8554	0.8818	1.0000	1.0000	1.0000	1.0000
Management	0.0193	0.1985	0.8165	0.5000	0.5191	0.6157	0.9134	0.8909	0.9316	0.9531	0.9919	1.0000
Overall	0.0355	0.2044	0.8203	0.7490	0.3430	0.4699	0.8834	0.9350	0.9487	0.9654	0.9928	0.9993

TABLE IX

BASILINE PERFORMANCE BY SOURCE DATASET (RECOMPUTED FROM THE RELEASED PREDICTIONS; NO RE-TRAINING). "MICRO" IS THE SAMPLE-WEIGHTED AVERAGE REPORTED IN THE ORIGINAL SUBMISSION AND IS DOMINATED BY VinDr-MAMMO; "MACRO" IS THE SOURCE-BALANCED AVERAGE REQUESTED BY REVIEWER 2.

Setting	Source	BLEU-4	ROUGE-L	CKR
Zero-shot	CBIS-DDSM	0.0421	0.2066	0.7371
	CMMD	0.0389	0.2106	0.7480
	INbreast	0.0187	0.1752	0.8000
	VinDr-Mammo	0.0259	0.1999	0.7626
	<i>micro</i>	0.0355	0.2044	0.7490
	<i>macro</i>	0.0314	0.1981	0.7619
3-shot	CBIS-DDSM	0.2341	0.3759	0.9049
	CMMD	0.2747	0.4173	0.9490
	INbreast	0.1975	0.3055	0.9250
	VinDr-Mammo	0.5328	0.6328	0.9695
	<i>micro</i>	0.3430	0.4699	0.9350
	<i>macro</i>	0.3098	0.4329	0.9371
QLoRA fine-tuned	CBIS-DDSM	0.9820	0.9889	1.0000
	CMMD	0.9519	0.9716	0.9961
	INbreast	0.9665	0.9798	1.0000
	VinDr-Mammo	0.9010	0.9297	1.0000
	<i>micro</i>	0.9487	0.9654	0.9993
	<i>macro</i>	0.9504	0.9675	0.9990

matched mammogram, a different patient's breast, with the question.

Condition (e), not (c), decides the matter: any mammogram may prompt a radiological register and a better score without a fact being read from the pixels; only the correct-versus-wrong contrast isolates information specific to *this* case. We define

$$\Delta_{\text{vis}} = \text{score}(c) - \text{score}(e) \quad (1)$$

and treat Δ_{vis} , not the raw score, as the measure of visual grounding: memorised templates raise BLEU without bound but cannot raise it. It therefore answers Reviewer 1's charge of template memorisation directly, and we report it throughout.

In partial information decomposition terms [49], (a) $\rightarrow$ (b) isolates language-unique information, (c) $-$ (e) visual-unique information, and (d) against (b) exposes whether the sources are redundant or synergistic.

Experiments use CBIS-DDSM, the only source supporting all nine question types, with full morphological descriptors, biopsy-confirmed pathology and genuinely abnormal cases (no BI-RADS 1; 59% BI-RADS 4–5). Restricting to it is conservative: VinDr-Mammo, 65% of MAXVQA, is dominated by BI-RADS 1 examinations, where a mismatched image resembles the correct one and would suppress Δ_{vis} for reasons unrelated to the model. Pairings were verified against the DICOM headers. We evaluate the open-weight general-purpose Qwen2-VL-7B; MedGemma, a domain-specific medical foundation model, is access-gated and could not be obtained within the revision window, though our harness supports it and it will be added once access is granted.

1) Off-the-shelf models do not read the mammogram:

Zero-shot, Qwen2-VL-7B gives an unambiguously negative result (Table X): $\Delta_{\text{vis}} = 0.0014$ in BLEU-4 (95% CI $[-0.0001, +0.0030]$, $p = 0.17$), no question type surviving correction for multiple comparisons. *The correct mammogram is statistically indistinguishable from a random one.* The structured findings, by contrast, cut the rate of claims contradicting the annotation six-fold, from 0.307 to 0.051 ($p = 6.6 \times 10^{-21}$): essentially all information the model exploits is language-unique, the visual-unique component indistinguishable from zero.

The mammogram does not merely fail to help, it *raises* the contradiction rate, from 0.307 to 0.356, and equally when the image is wrong (0.351). Rather than abstain, the model asserts BI-RADS categories, laterality and morphology conflicting with the ground truth: confident fabrication in radiological vocabulary, a clinically hazardous failure invisible to any evaluation without a mismatched-image control.

2) *A model trained on the images does read them:* The zero-shot result licenses no conclusion about the dataset; the question is whether a *trained* model can use these images. We fine-tuned Qwen2-VL-7B with 4-bit QLoRA on up to 8,000 CBIS examples using *only the image and the question*, withholding the findings throughout training, so the pixels are the model's sole route to the answer.

The mismatched-image control now separates cleanly (Table X): at 8,000 examples, $\Delta_{\text{vis}} = 0.087$ in BLEU-4 (95% CI $[+0.047, +0.126]$, $p = 4.4 \times 10^{-6}$), with concordant

TABLE X

FIVE-CONDITION IMAGE-ABLATION LADDER, ZERO-SHOT AND AFTER FINE-TUNING ON 8,000 CBIS EXAMPLES WITH THE IMAGE AND QUESTION ONLY. “CONTRA.” IS THE RATE OF CLAIMS CONTRADICTING THE ANNOTATION. ZERO-SHOT, THE CORRECT AND MISMATCHED MAMMOGRAMS ARE STATISTICALLY INDISTINGUISHABLE; AFTER FINE-TUNING A SIGNIFICANT GAP OPENS.

Condition	Zero-shot		Fine-tuned
	BLEU-4	Contra.	BLEU-4
(a) question only	0.0211	0.307	—
(b) question + findings	0.0250	0.051	—
(c) image + question	0.0223	0.356	0.602
(d) image + question + findings	0.0256	0.051	—
(e) <i>mismatched</i> image + question	0.0209	0.351	0.515
$\Delta_{\text{vis}} = (c) - (e)$	0.0014		0.087
95% CI	[-0.0001, +0.0030]		[+0.047, +0.126]
p-value	0.17 (n.s.)		4.4×10^{-6}
ROUGE-L/BERTScore gap			+0.069/+0.072

gains in ROUGE-L (0.069) and BERTScore (0.072). A model trained on MAXVQA extracts case-specific information from the mammogram.

3) *The image recovers a verifiable clinical fact*: BLEU-4 understates the effect: of the roughly sixty tokens in a reference answer only a handful carry case-specific content, the density category, the BI-RADS number, one margin descriptor, the rest being scaffolding shared across every case of a type. We therefore score the *clinical fact* asserted rather than its wording, parsing each answer’s attribute value and checking it against the annotation; this also adds the language-only baseline (a) that Reviewer 2 requested and our earlier (c)-versus-(e) runs omitted.

Breast density is unambiguous (Table XI). The model names the correct ACR category in 66.1% of cases with the correct mammogram, against 25.9% with no image, 33.0% with a mismatched one, and a majority-class rate of 41.1%: 40 points over the language-only baseline ($p = 2.2 \times 10^{-9}$) and 33 points over the mismatched control ($p = 1.2 \times 10^{-6}$). BLEU orders them identically, the mismatched image scoring *below* the no-image baseline (0.521 against 0.542, correct image 0.622): a wrong mammogram degrades the answer relative to no image at all.

We are equally clear about where this does *not* hold. For pathology (benign vs. malignant) the image lifts accuracy from 32.4% to 42.6%, still *below* the 54.1% majority-class rate: the model has not learned to call malignancy, and the gain over the mismatched control is not significant. For abnormality type (mass vs. calcification) it adds essentially nothing over the language-only baseline (55.9% $\rightarrow$ 59.0%). Our claim is specific: a model trained on MAXVQA recovers breast density from the mammogram but does not thereby become a diagnostic system, and we do not present it as one.

4) *Visual grounding is acquired, not memorised*: Were the gain template memorisation, it would be present at the smallest training budget and would not grow. Visual reading must instead be acquired, and should emerge with more mammograms; Fig. 2 separates the accounts.

Across 500, 1,000, 2,000, 4,000 and 8,000 examples, Δ_{vis} grows from 0.014 to 0.087 (Spearman $\rho = 0.900$, $p = 0.037$),

TABLE XI

ATTRIBUTE RECOVERY: ACCURACY AT WHICH THE FINE-TUNED MODEL ASSERTS THE CORRECT ATTRIBUTE VALUE, AT 896 PX. (A) IS THE LANGUAGE-ONLY BASELINE; (E) IS A MISMATCHED PATIENT’S MAMMOGRAM; “MAJ.” IS THE MAJORITY-CLASS RATE.

Attribute	<i>n</i>	maj.	(a) no img	(c) img	(e) wrong	<i>c - a</i>
Breast density (4-cl.)	112	41.1%	25.9%	66.1%	33.0%	+40.2
Pathology (2-cl.)	444	54.1%	32.4%	42.6%	37.4%	+10.1
Abnormality (2-cl.)	329	59.0%	55.9%	59.0%	50.5%	+3.0

not significant at 500 or 1,000 and becoming so only at 2,000: visual grounding is absent at small scale and appears with data. A memorising model cannot produce an *emerging* Δ_{vis} , memorisation being indifferent to which image is supplied: this is the strongest evidence we have against the memorisation charge.

Both curves rise, the correct-image score from 0.479 to 0.602 and the mismatched-image score from 0.465 to 0.515; the gap opens because the former climbs faster, not because the latter stops. A model given a wrong mammogram keeps improving on answer templates and question–answer priors, and we claim no template ceiling.

The curve saturates, and this bounds our claim: Doubling the training set from 4,000 to 8,000 examples increased Δ_{vis} by 0.003, against 0.037 and 0.044 for the two preceding doublings: visual grounding on MAXVQA is near its asymptote at 8,000 examples, and further QA pairs should not raise it materially. Sec. IV-F.5 identifies the binding constraint instead.

Two results temper the finding. First, the correct image improves *wording overlap* with the reference but not clinical faithfulness relative to using no image: the contradiction rate is 0.466 with the correct mammogram and 0.479 with none, a difference that is not significant. A *mismatched* image does raise it (0.537, $p < 10^{-5}$ against the correct image), so the model is demonstrably sensitive to which mammogram it sees, but as closer reference matching, not as fewer false statements than with no image. Second, on counterfactual questions Δ_{vis} is *negative* (−0.029 at 448 px), the correct image making the answer slightly worse. A model trained on MAXVQA extracts case-specific information from the mammogram but does not thereby become clinically reliable, and we make no such claim.

5) *Where the image can help, and where it cannot*: The gain is unevenly distributed across question types (Table XII). On the three feature-significance types, which draw on five, five and six distinct answers respectively (Sec. IV-B), Δ_{vis} is *exactly* 0.000 at every training scale and resolution tested.

On these types the fine-tuned model attains a BLEU-4 of 1.0000, a *perfect* reproduction of the reference, under the correct image *and* under a mismatched one, emitting byte-identical strings for 53/53 items. Having learned 4 – 5 answer strings, it succeeds completely without the mammogram.

The question gives the answer away: in 91% of feature-significance items it *names the feature it is asking about*, “Why are coarse heterogeneous calcifications clinically significant?”, making the answer a deterministic function of the question text. This is the reviewers’ central criticism, shown here more sharply than it was put to us: the BLEU-4 of 1.0000 on these

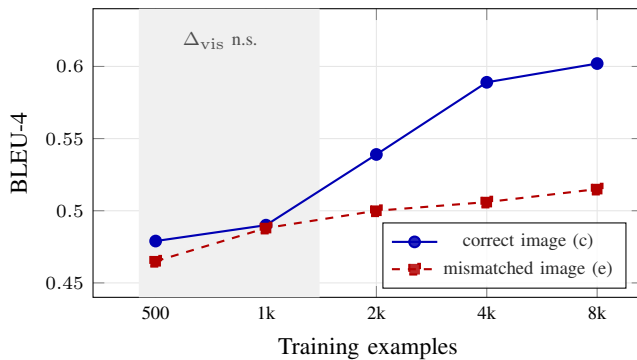


Fig. 2. Visual grounding is *acquired*, not memorised. Δ_{vis} , the vertical distance between the curves, is indistinguishable from zero at 500 and 1,000 training examples (shaded region) and becomes significant only at 2,000, reaching 0.087 at 8,000 (Spearman $\rho = 0.900$, $p = 0.037$); zero-shot it is 0.0014 (n.s.). Interpretation in Sec. IV-F.5.

TABLE XII

IMAGE SENSITIVITY Δ_{vis} (BLEU-4) BY QUESTION TYPE AT $n = 8,000$, AT TWO INPUT RESOLUTIONS. “UNIQ.” IS THE NUMBER OF DISTINCT ANSWER STRINGS AND H THE ANSWER ENTROPY. RESOLUTION LEAVES THE THREE DEGENERATE TYPES UNCHANGED (SEE TEXT).

Question type	Uniq.	H	Δ_{vis} 448	Δ_{vis} 896
Density explanation	69	5.30	0.211	0.206
Descriptive	2,486	11.02	0.124	0.153
Differential diagnosis	20	3.52	0.057	0.112
Assessment reasoning	808	8.60	0.098	0.089
Management	52	5.32	0.102	0.079
Counterfactual	56	4.07	-0.029	0.058
<i>Degenerate: BLEU-4 = 1.0000 with the WRONG image</i>				
Feat. sig. (margins)	5	2.22	0.000	0.000
Feat. sig. (shape)	5	2.13	0.000	0.000
Feat. sig. (calcif.)	6	1.97	0.000	0.000
Overall			0.087	0.100

rows in our original submission was a model reproducing five strings.

We tested the alternative explanation and it is wrong: These three types depend on fine spatial detail, margin spiculation, mass shape, calcification morphology, the last involving structures of 0.1–0.5 mm, while our encoder received 448×448 inputs, at which one pixel subtends roughly 0.5 mm of breast. Microcalcifications lie at or below that limit, so a resolution artefact was a live possibility.

We repeated the 8,000-example fine-tune at 896×896 (1,024 visual tokens against 256), holding data, seed, budget and merger adaptation fixed. The prediction, registered beforehand, that the three types would rise above zero was *wrong*: all three remain at exactly 0.000. Quadrupling the visual token budget changes nothing, because there is nothing in the image the model needs; resolution is not the explanation, the corpus is.

6) Resolution helps, but only where the answer varies: The same ablation separates the non-degenerate types. At 896×896 , overall image sensitivity rises from 0.087 to 0.100 (95% CI $[-0.060, +0.141]$, $p = 4.4 \times 10^{-8}$), concentrated where the label carries variance: differential diagnosis from

0.057 to 0.112, descriptive characterisation from 0.124 to 0.153, counterfactual reasoning changing sign from -0.029 to 0.058. Density explanation, a global low-spatial-frequency property never resolution-limited, is essentially unchanged ($0.211 \rightarrow 0.206$), as we predicted in advance.

The two conclusions point opposite ways. Resolution *does* matter here, and our original 448-pixel budget understated what a model can extract from a mammogram; but it cannot rescue a question whose answer is fixed by its own wording, and on the degenerate types it buys nothing.

7) Consequences for dataset design: Two transferable conclusions survive. First, *the number of QA pairs is not the binding constraint*: the curve saturates, the 4,000-to-8,000 doubling gaining only 0.003 in Δ_{vis} . Templating structured annotations gives a corpus a ceiling, set by the entropy of its answer distribution, on the visual grounding it can teach: six distinct answers cannot teach a model to see, however many pairs a question type contains. Second, *a corpus must preserve the spatial resolution of the findings it asks about*: a benchmark posing questions about 0.3 mm calcifications while distributing images that cannot resolve them is not testing what it claims to test. We under-specified this in our original release and correct it here, shipping the DICOM-to-image pipeline with the resolution at which each result was obtained.

G. Leave-One-Dataset-Out Transfer

Reviewer 2 asked whether the benchmark supports generalisable reasoning or merely source-specific pattern matching, and proposed a leave-one-dataset-out (LODO) test. We ran it for all four sources, training on three and testing on the held-out fourth with the same QLoRA recipe. Mean held-out BLEU-4 is 0.77 (range 0.695–0.860), which looks like strong transfer to unseen annotation protocols. The per-question-type breakdown (Table XIII) shows why that reading is misleading. Transfer is near-perfect precisely on the types whose answers are shared templates, the three feature-significance types reaching 1.000 on a source held out of training because those five or six answer strings are identical across sources, and collapses where the answer carries source-specific content, differential diagnosis on held-out INbreast scoring 0.039 and assessment reasoning on held-out CBIS-DDSM 0.386. LODO thus independently corroborates the redundancy audit (Sec. IV-B) and the vision ablation (Sec. IV-F). We report it per source and per question type and draw no stronger conclusion.

H. Planned Extensions

The evaluation above establishes visual grounding on an objective, image-derived attribute using a non-circular protocol (Sec. IV-F.3). Four complementary evaluations follow naturally, set out here so the boundary of the present claims is unambiguous. First, an independent test set of clinician-authored *free-text* answers would let explanation quality be scored against human references rather than ground-truth labels; this is the strongest form of the non-templated evaluation and we are actively pursuing it. Second, a leave-one-dataset-out protocol, training on three sources and testing on the fourth, would separate generalisable reasoning from source-specific

TABLE XIII

LEAVE-ONE-DATASET-OUT TRANSFER: BLEU-4 ON THE HELD-OUT SOURCE AFTER TRAINING ON THE OTHER THREE, BY QUESTION TYPE. A DASH MARKS A QUESTION TYPE NOT PRESENT IN THAT SOURCE. SOURCE: PER-HOLDOUT `FINETUNED_METRICS.JSON`.

Held-out source (test)	CBIS	CMMD	INbreast	VinDr
Feat. sig. (calcif.)	0.88	1.00	–	–
Feat. sig. (margins)	1.00	1.00	–	–
Feat. sig. (shape)	0.91	1.00	–	–
Density explanation	0.75	0.94	0.93	0.95
Counterfactual	0.88	0.91	–	–
Management	0.61	0.89	0.91	0.71
Descriptive	0.66	0.65	0.82	0.68
Assessment reasoning	0.39	0.74	0.77	0.76
Differential	0.52	0.60	0.04	–
Overall	0.74	0.86	0.70	0.78

pattern matching. Third, an item-level agreement study, with a multi-family judge panel and the human raters scoring an identical item set, would place LLM–human concordance on a firmer footing than the present near-disjoint samples allow. Fourth, component ablations of the generation pipeline (the ROUGE-L similarity filter, the BI-RADS knowledge base, and the paraphrasing module) would quantify each element’s contribution to answer quality and diversity. None is required to support the claims we make here; each would extend them.

V. BIAS, FAIRNESS, AND DEPLOYMENT CONSIDERATIONS

MAXVQA inherits the composition of its four sources, and the resulting skews bear on how models trained on it should, and should not, be used. *Geographic and demographic coverage* spans North American (CBIS-DDSM), European (INbreast), Southeast Asian (VinDr-Mammo) and East Asian (TOMPEI-CMMD) screening populations, but unevenly: VinDr-Mammo alone contributes 65.2% of annotations, so aggregate metrics are dominated by a single Southeast Asian cohort, and African and South American populations are absent entirely. *Acquisition era and modality* are mixed, CBIS-DDSM being digitised screen-film and the other three full-field digital, so image characteristics are confounded with source and, through it, with population. *Label distributions* are skewed: TOMPEI-CMMD is heavily malignant-enriched (87% of its cases) whereas VinDr-Mammo is dominated by BI-RADS 1–2 normals, so any model trained on the pooled corpus absorbs these priors. One that learns “predict benign” or “predict the modal density” can post strong aggregate numbers while failing the minority it most needs to serve; the per-source and per-question-type reporting we adopt throughout (Secs. IV-F, IV-G) makes such failures visible rather than letting them average out.

We therefore frame MAXVQA as a research resource, not a clinical tool. Any model trained on it must be evaluated across demographic subgroups, acquisition modalities and BI-RADS strata before deployment, and would require prospective external validation and regulatory clearance (including FDA clearance for clinical decision support) before use in

patient care. The dataset suits developing and benchmarking explanatory methods; it is not a substitute for representative, prospectively collected clinical data.

VI. CONCLUSION

MAXVQA is a large-scale dataset of 125,591 explanatory question–answer pairs for mammography VQA, built from four complementary source databases, CBIS-DDSM, VinDr-Mammo, INbreast, and TOMPEI-CMMD, spanning three continents. It addresses a critical and underserved gap: where existing VQA benchmarks elicit single-word classifications, MAXVQA demands multi-sentence explanations articulating feature significance, counterfactual reasoning, differential diagnosis, and management recommendations, grounded throughout in ACR BI-RADS Atlas standards and FDA MQSA requirements. This reflects our core argument, that closing the gap between AI capabilities and clinical practice requires not only better models but better data, encoding the structured, explanatory reasoning radiologists employ at each stage of mammographic interpretation.

The evidence we can offer for the dataset’s clinical quality is narrower than we originally claimed, and we prefer to be precise about its limits. Two senior consultant radiologists rated 94 QA pairs across four dimensions; their scores do not agree beyond chance on clinical accuracy, completeness or clinical relevance (Spearman $\rho = 0.04$ – 0.14 , all non-significant), because one rater assigned the maximum score to 96% of items. The evaluation is therefore best characterised as a preliminary expert spot-check resting effectively on a single discriminating rater, not as validation. The multi-agent LLM evaluation, originally presented as corroborating it, cannot serve that purpose either: the 8B judge confined its accuracy scores to $\{4, 5\}$ across all 600 judgements, and the two evaluations were run on almost disjoint item sets. We retain it only as a scalable secondary screen, and report per-dimension inter-agent κ (accuracy 0.560, completeness 0.272, relevance 0.487) in place of the composite quoted originally, which corresponded to no single statistic.

The text-only baselines establish a three-point curve (zero-shot BLEU-4 0.036, 3-shot 0.343, QLoRA 0.949) which we now interpret as measuring acquisition of an explanation style rather than of clinical reasoning, the references being templates (Sec. IV-B); the saturated rows are a property of the reference set, not an achievement of the model. What the curve does show, and what remains a genuine contribution, is that a schema-grounded explanatory style can be induced very cheaply, three examples recovering most of the form and a thousand nearly all of it, a useful fact for anyone building explanation-generating systems on structured radiological findings.

Several limitations should inform how the dataset is used and extended. The pipeline is template-based rather than clinician-authored, ensuring consistency and regulatory alignment at the cost of answer diversity: the 125,591 QA pairs contain 5,163 distinct answers, and on the structured question types, which draw on five or six answer strings, the near-perfect fine-tuning scores reflect template reproduction rather

than clinical reasoning. A related consequence, discussed in Sec. III-B.5, is that the principal error mode is not hallucination but correlated systematic error: a single incorrect knowledge-base entry propagates identically across every pair it touches. The baselines in the original submission were text-only, supplying clinical metadata as structured input without mammographic images. Because the answers are deterministic functions of that metadata (Sec. IV-B), those experiments could not in principle have demonstrated visual grounding, whatever their scores. Sec. IV-F therefore reports vision-grounded baselines under a controlled ablation ladder including a mismatched-image control, and we regard the outcome of that control, not the metric values, as the test of whether this resource merits the term “VQA”. The paraphrasing module, though implemented, has not yet been empirically evaluated for its effect on downstream generalization and warrants dedicated investigation. Finally, although the four sources span North American, European, Southeast Asian, and East Asian screening populations, models trained on MAXVQA should be evaluated across demographic groups before clinical deployment. Future work should extend the dataset toward multi-view reasoning across CC and MLO projections, temporal reasoning for sequential examinations, and integration with larger-scale collections. User studies of how radiologists interact with explanatory VQA outputs in actual workflows will ultimately determine whether the reasoning structure encoded in MAXVQA translates into measurable gains in diagnostic confidence and efficiency.

ACKNOWLEDGMENT

We thank the creators of CBIS-DDSM, VinDr-Mammo, INbreast, and TOMPEI-CMMD for making their datasets publicly available.

DATA AVAILABILITY

All datasets used in this investigation are publicly available. MAXVQA, the generation pipeline and the evaluation code, together with the released train/validation/test splits and the result files underlying the tables reported here, are available at https://github.com/Minds-R-Lab/VQA_Mammo. All source datasets are publicly available under appropriate licenses: CBIS-DDSM under CC BY 3.0, VinDr-Mammo under PhysioNet Credentialed Health Data License 1.5.0, INbreast under a research use agreement, and TOMPEI-CMMD under The Cancer Imaging Archive (TCIA) Data Usage Policy.

MAXVQA is intended for research and educational purposes. Clinical deployment would require rigorous validation, regulatory approval (including FDA clearance for clinical decision support software), and appropriate human oversight.

[1] H. Sung, J. Ferlay, R. L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, and F. Bray, “Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries,” *CA: A Cancer Journal for Clinicians*, vol. 71, no. 3, pp. 209–249, 2021.

[2] Y. Zafari, S. Soliman, E. A. Rahed, and M. Mabrok, “The need to move beyond explainability toward chain-of-thought reasoning: A focus on ai for mammography,” in *AAAI Bridge Program on AI for Medicine and Healthcare*. PMLR, 2026, pp. 322–329.

[3] S. M. McKinney, M. Sieniek, V. Godbole, J. Godwin, N. Antropova, H. Ashrafi, T. Back, M. Chesus, G. S. Corrado, A. Darzi, M. Etemadi, F. Garcia-Vicente, F. J. Gilbert, M. Halling-Brown, D. Hassabis, S. Jansen, A. Karthikesalingam, C. J. Kelly, D. King, J. R. Ledsam, D. Melnick, H. Mostofi, L. Peng, J. J. Reicher, B. Romera-Paredes, R. Sidebottom, M. Suleyman, D. Tse, K. C. Young, J. De Fauw, and S. Shetty, “International evaluation of an AI system for breast cancer screening,” *Nature*, vol. 577, no. 7788, pp. 89–94, 2020.

[4] A. Rodríguez-Ruiz, E. Krupinski, J.-J. Mordang, K. Schilling, S. H. Heywang-Köbrunner, I. Sechopoulos, and R. M. Mann, “Detection of breast cancer with mammography: Effect of an artificial intelligence support system,” *Radiology*, vol. 290, no. 2, pp. 305–314, 2019.

[5] Y. Zafari, R. Elalfy, M. Nouman, S. Al-Maadeed, T. Khattab, E. A. Rashed, and M. Mabrok, “Multi-modal deep learning in breast cancer diagnosis: A review of recent advances,” in *2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA)*. IEEE, 2025, pp. 1–6.

[6] Y. Zafari, R. Elalfy, M. Mabrok, S. Al-Maadeed, T. Khattab, and E. A. Rashed, “A hybrid cnn-vssm model for multi-view, multi-task mammography analysis: Robust diagnosis with attention-based fusion,” *arXiv preprint arXiv:2507.16955*, 2025.

[7] Y. Du, L. Chen, and N. C. Dvornek, “Glam: Geometry-guided local alignment for multi-view vlp in mammography,” *arXiv preprint arXiv:2509.10344*, 2025.

[8] S.-F. Zheng, H. Lee, T. Kooi, and A. Diba, “Mv-mlm: Bridging multi-view mammography and language for breast cancer diagnosis and risk prediction,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 1213–1222.

[9] S. Ghosh, C. B. Poynton, S. Visweswaran, and K. Batmanghelich, “Mammo-clip: A vision language foundation model to enhance data efficiency and robustness in mammography,” in *International conference on medical image computing and computer-assisted intervention*. Springer, 2024, pp. 632–642.

[10] J. Zhu, F. Huang, Q. Luo, and H. Chen, “A benchmark for breast cancer screening and diagnosis in mammogram visual question answering,” *Nature Communications*, vol. 16, p. 11683, 2025.

[11] Y. Zafari, H. Pan, G. Durak, U. Bagci, E. A. Rashed, and M. Mabrok, “Mammoclean: Toward reproducible and bias-aware ai in mammography through dataset harmonization,” *IEEE Access*, vol. 14, pp. 93 834–93 853, 2026.

[12] X. Liu, L. Sun, C. Li, B. Han, W. Jiang, T. Yuan, W. Liu, Z. Liu, Z. Yu, and B. Liu, “Lesion asymmetry screening assisted global awareness multi-view network for mammogram classification,” *IEEE Transactions on Medical Imaging*, 2025.

[13] G. Van Tulder, Y. Tong, and E. Marchiori, “Multi-view analysis of unregistered medical images using cross-view transformers,” in *Medical Image Computing and Computer Assisted Intervention—MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part III* 24. Springer, 2021, pp. 104–113.

[14] M. Tardy and D. Mateus, “Looking for abnormalities in mammograms with self-and weakly supervised reconstruction,” *IEEE Transactions on Medical Imaging*, vol. 40, no. 10, pp. 2711–2722, 2021.

[15] F. Manigrasso, R. Milazzo, A. S. Russo, F. Lamberti, F. Strand, A. Pagnani, and L. Morra, “Mammography classification with multi-view deep learning techniques: Investigating graph and transformer-based architectures,” *Medical Image Analysis*, vol. 99, p. 103320, 2025.

[16] X. Chen, Y. Li, M. Hu, E. Salari, X. Chen, R. L. Qiu, B. Zheng, and X. Yang, “Mammo-clip: Leveraging contrastive language-image pre-training (clip) for enhanced breast cancer diagnosis with multi-view mammography,” *arXiv preprint arXiv:2404.15946*, 2024.

[17] Z. Zhang, Y. Xie, F. Xing, M. McGough, and L. Yang, “MDNet: A semantically and visually interpretable medical image diagnosis network,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 6428–6436.

[18] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-cam: Visual explanations from deep networks via gradient-based localization,” in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 618–626.

[19] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[20] N. Shifa, M. Saleh, Y. Akbari, and S. Al Maadeed, “A review of

- explainable ai techniques and their evaluation in mammography for breast cancer screening,” *Clinical Imaging*, p. 110492, 2025.
- [21] C. Rudin, “Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead,” *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
 - [22] N. Arun, N. Gaw, P. Singh, K. Chang, M. Aggarwal, B. Chen, K. Hoebel, S. Gupta, J. Patel, M. Gidwani *et al.*, “Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging,” *Radiology: Artificial Intelligence*, vol. 3, no. 6, p. e200267, 2021.
 - [23] A. J. DeGrave, J. D. Janizek, and S.-I. Lee, “Ai for radiographic covid-19 detection selects shortcuts over signal,” *Nature Machine Intelligence*, vol. 3, no. 7, pp. 610–619, 2021.
 - [24] I. Banerjee, K. Bhattacharjee, J. L. Burns, H. Trivedi, S. Purkayastha, L. Seyyed-Kalantari, B. N. Patel, R. Shiradkar, and J. Gichoya, “‘shortcuts’ causing bias in radiology artificial intelligence: causes, evaluation, and mitigation,” *Journal of the American College of Radiology*, vol. 20, no. 9, pp. 842–851, 2023.
 - [25] A. J. Barnett, F. R. Schwartz, C. Tao, C. Chen, Y. Ren, J. Y. Lo, and C. Rudin, “A case-based interpretable deep learning model for classification of mass lesions in digital mammography,” *Nature Machine Intelligence*, vol. 3, no. 12, pp. 1061–1070, 2021.
 - [26] B. Kim, M. Wattenberg, J. Gilmer, C. Cai, J. Wexler, F. Viegas, and R. Sayres, “Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV),” in *International Conference on Machine Learning (ICML)*, 2018, pp. 2668–2677.
 - [27] T. Van Sonsbeek and M. Worring, “Case-based reasoning for medical visual question answering,” in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, 2022, pp. 572–582.
 - [28] B. Jing, P. Xie, and E. Xing, “On the automatic generation of medical imaging reports,” in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 2577–2586.
 - [29] C. Li, Y. Zhang, C. Xu, Z. Song, and Z. Tu, “Dynamic graph enhanced contrastive learning for chest X-ray report generation,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 3334–3343.
 - [30] J. J. Lau, S. Gayen, A. Ben Abacha, and D. Demner-Fushman, “A dataset of clinically generated visual questions and answers about radiology images,” *Scientific Data*, vol. 5, p. 180251, 2018.
 - [31] A. Ben Abacha, S. A. Hasan, V. V. Datla, J. Liu, D. Demner-Fushman, and H. Müller, “VQA-Med: Overview of the medical visual question answering task at ImageCLEF 2019,” in *CLEF 2019 Working Notes*, ser. CEUR Workshop Proceedings, vol. 2380, 2019.
 - [32] B. Liu, L.-M. Zhan, L. Xu, L. Ma, Y. Yang, and X.-M. Wu, “SLAKE: A semantically-labeled knowledge-enhanced dataset for medical visual question answering,” in *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2021, pp. 1650–1654.
 - [33] X. He, Y. Zhang, L. Mou, E. Xing, and P. Xie, “PathVQA: 30000+ questions for medical visual question answering,” *arXiv preprint arXiv:2003.10286*, 2020.
 - [34] X. Zhang, C. Wu, Z. Zhao, W. Lin, Y. Zhang, Y. Wang, and W. Xie, “PMC-VQA: Visual instruction tuning for medical visual question answering,” *arXiv preprint arXiv:2305.10415*, 2023.
 - [35] D. V. M. Nguyen, C. T. Truong, P. H. Tran, H. T. Le, N. V.-T. Dat, T. H. Pham, and K. V. Nguyen, “ViX-Ray: A Vietnamese chest X-ray dataset for vision-language models,” in *Proc. Int. Conf. Language Resources and Evaluation (LREC)*, 2026, arXiv:2603.15513.
 - [36] R. S. Lee, F. Gimenez, A. Hoogi, K. K. Miyake, M. Gorovoy, and D. L. Rubin, “A curated mammography data set for use in computer-aided detection and diagnosis research,” *Scientific Data*, vol. 4, p. 170177, 2017.
 - [37] H. T. Nguyen, H. Q. Nguyen, H. H. Pham, K. Lam, L. T. Le, M. Dao, and V. Vu, “VinDr-Mammo: A large-scale benchmark dataset for computer-aided detection and diagnosis in full-field digital mammography,” *Scientific Data*, vol. 10, p. 277, 2023.
 - [38] I. C. Moreira, I. Amaral, I. Domingues, A. Cardoso, M. J. Cardoso, and J. S. Cardoso, “INbreast: Toward a full-field digital mammographic database,” *Academic Radiology*, vol. 19, no. 2, pp. 236–248, 2012.
 - [39] C. Cui, L. Li, H. Cai, Z. Fan, L. Zhang, T. Dan, J. Li, and J. Wang, “The Chinese mammography database (CMMD): An online mammography database with biopsy confirmed types for machine diagnosis of breast,” *The Cancer Imaging Archive*, 2021.
 - [40] Y. Kashiwada, E. Takaya, M. Hiroya, N. Matsuda, T. Yashima, T. Kobayashi, G. Tamiya, and T. Ueda, “Tompei-cmmd dataset (version 1) [dataset],” *The Cancer Imaging Archive*, 2025.
 - [41] E. A. Sickles, C. J. D’Orsi, L. W. Bassett, C. M. Appleton, W. A. Berg, and E. S. Burnside, *ACR BI-RADS Atlas: Breast Imaging Reporting and Data System*, 5th ed. American College of Radiology, 2013.
 - [42] U.S. Food and Drug Administration, “Mammography quality standards act (MQSA) regulations,” 2024, accessed: 2024-12-01. [Online]. Available: <https://www.fda.gov/radiation-emitting-products/mammography-quality-standards-act-mqsa-and-mqsa-program/regulations-mqsa>
 - [43] X. Tang, A. Zou, Z. Zhang, Y. Zhao, X. Zhang, A. Cohan, and M. Gerstein, “MedAgents: Large language models as collaborators for zero-shot medical reasoning,” *arXiv preprint arXiv:2311.10537*, 2023.
 - [44] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten *et al.*, “The Llama 3 herd of models,” *arXiv preprint arXiv:2407.21783*, 2024.
 - [45] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized language models,” in *Advances in Neural Information Processing Systems*, vol. 36, 2024.
 - [46] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “BLEU: A method for automatic evaluation of machine translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2002, pp. 311–318.
 - [47] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *International Conference on Learning Representations (ICLR)*, 2020.
 - [48] J. R. Landis and G. G. Koch, “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
 - [49] W. Fang, T. Zhang, W. Tao, and A. Chan, “Towards understanding modality interaction in multimodal language models via partial information decomposition,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2026, arXiv:2606.00959.

APPENDIX

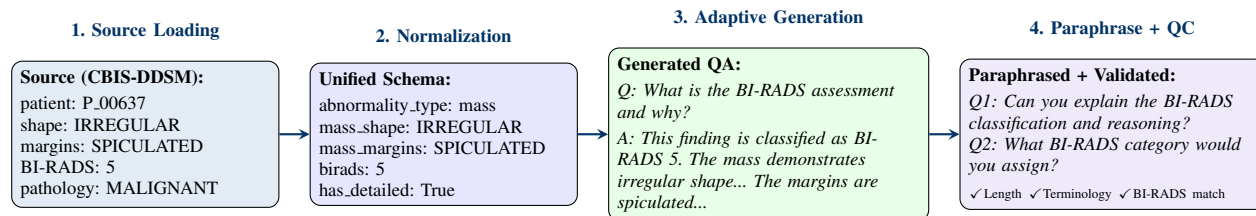


Fig. 3. End-to-end walkthrough: a CBIS-DDSM annotation with irregular shape and spiculated margins is normalized, used to generate an assessment-reasoning QA pair from the BI-RADS knowledge base, then paraphrased and validated. The annotation yields 7 QA pairs, one per applicable question type.

TABLE XIV
REPRESENTATIVE QA PAIRS FROM MAXVQA ILLUSTRATING EACH QUESTION TYPE. ALL EXAMPLES ARE DRAWN FROM CBIS-DDSM ANNOTATIONS WITH BIOPSY-CONFIRMED PATHOLOGY.

Question Type	Question	Answer (abridged)
Assessment Reasoning	What is the BI-RADS assessment and why?	This finding is classified as BI-RADS 5 (Highly Suggestive of Malignancy). The mass demonstrates irregular shape, indicating uncontrolled asymmetric growth characteristic of malignancy. The margins are spiculated, representing either direct tumor extension along Cooper's ligaments or desmoplastic reaction. This margin type has a PPV of $\approx 85\%$ for malignancy. BI-RADS 5 indicates $>95\%$ likelihood of cancer; tissue diagnosis via image-guided biopsy is required. Pathology confirmed: malignant.
Counterfactual	Why is this not a cyst?	This lesion cannot be classified as a cyst because it lacks the characteristic features. Simple cysts typically appear oval or round with circumscribed margins. In this case, the mass has irregular shape (typical cyst: oval/round) and spiculated margins (typical cyst: circumscribed). These features are inconsistent with a simple cyst and raise suspicion for malignancy.
Feature Significance	What is the significance of the spiculated margins?	Spiculated margins demonstrate thin lines radiating from the mass into surrounding tissue, creating a stellate appearance. These spiculations represent either direct tumor extension along Cooper's ligaments and fibrous septa, or a desmoplastic reaction induced by the tumor. This margin type has a PPV of $\approx 85\%$ for malignancy, making it one of the most important predictors of breast cancer in mammography.
Management	What is the recommended management for this finding?	Based on the BI-RADS 4 assessment, this finding is suspicious. Tissue diagnosis via image-guided biopsy is recommended. The probability of malignancy in BI-RADS 4 ranges from 2% to 95%, subdivided into 4A (low suspicion), 4B (moderate), and 4C (high suspicion). Core needle biopsy with imaging guidance is the preferred approach for histologic diagnosis.
Differential Diagnosis	What are the possible differential diagnoses?	Given the irregular shape and spiculated margins, the primary differential includes invasive ductal carcinoma (most common, $\approx 70\text{--}80\%$ of breast cancers), invasive lobular carcinoma, and tubular carcinoma. Less likely but possible: radial scar or complex sclerosing lesion, which can mimic spiculated masses mammographically. Tissue sampling is required for definitive diagnosis.
Descriptive	Describe the mammographic finding.	The mammographic finding is a mass with irregular shape and spiculated margins. The mass is located in the left breast. The irregular shape indicates asymmetric growth that does not conform to the expected contours of a benign process. The spiculated margins demonstrate radiating lines extending into the surrounding parenchyma.
Density Explanation	How does breast density affect interpretation?	The breast tissue is classified as ACR density category C (heterogeneously dense). Dense fibroglandular tissue can obscure lesions on mammography because masses and dense tissue share similar X-ray attenuation. The sensitivity of mammography decreases from $\approx 85\%$ in fatty breasts to $\approx 65\%$ in extremely dense breasts. Supplemental screening may be considered per clinical guidelines.